

Indian Institute of
Technology Kharagpur

FINAL EVALUATION

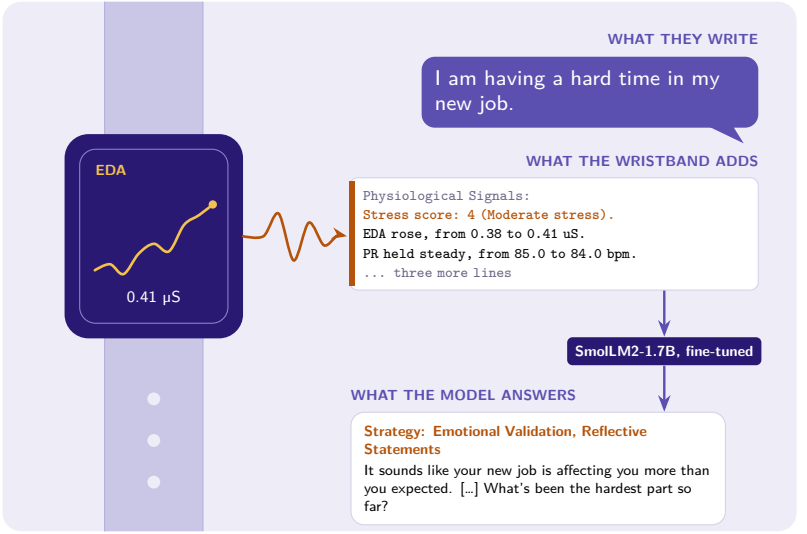
PhysioBase: Physiology-Aware Conversational Support

An emotional support model that reads a wristband's signal summary before it answers

Ashutosh Sharma
B. Siddaarth
Krishna Chaitanya
Nithish Kumar Mahesepu

Mentor: Sudipa Garain

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Why a support chatbot should read the body as well as the words



Widely used

Chatbots already help many people with well-being and stress management.



Text is not enough

Existing systems rely only on text, which limits how well they understand a person's true state.



Emotions are multimodal

They show in what we write, in speech and in heart rate variability.



The body is evidence

Physiological signals give objective, real-time indicators of stress.

So PhysioBase reads a short summary of a wristband's signals beside each message before it replies.

A text-only chatbot answers the words, not the person



1 A.M., THE NIGHT BEFORE HER FINAL EXAM

Riya says she is “fine”. Her body disagrees.

She sends a short, calm message. Her wristband records a different night, and a text-only chatbot only ever sees the message.

WHAT A TEXT-ONLY CHATBOT SEES

Riya: I’m fine, just tired.

Chatbot: Glad to hear it! Get some rest.

WHAT HER SMARTWATCH SEES

↑ Heart rate climbing

↓ Heart rate variability dropping

Both point to stress that the message never mentions.

REQUIREMENT

A multimodal chatbot that fuses what a person writes with their physiological signals.

22 participants, each with a wristband for a week

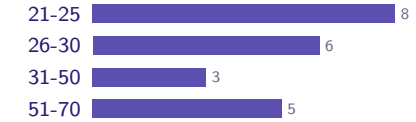
22
participants

13 / 9
women / men

21-67
years old, median 27

7 days
each wore the Embrace Plus, one support chat a day

AGE



OCCUPATION

PhD 4, BTech 3, MS 3, business 3, school teacher 2, housewife 2; BSc, MTech, nurse, home tutor and banker 1 each

#	Age	Gender	Profession
0	25	M	MS
1	27	F	Nurse
2	27	F	PhD
3	31	F	Home tutor
4	27	F	PhD
5	25	F	PhD
6	46	F	School teacher
7	21	M	BTech
8	21	F	BSc
9	22	M	MS
10	28	F	School teacher

#	Age	Gender	Profession
11	62	F	Housewife
12	21	M	BTech
13	22	F	MTech
14	27	M	Banker
15	21	M	BTech
16	33	F	PhD
17	26	F	MS
18	67	M	Businessman
19	65	M	Businessman
20	52	M	Businessman
21	57	F	Housewife

Four of six channels differ from the human range on the wrist

NEW SINCE MID-TERM

THE SIX CHANNELS

EDA, μS
sweat on the skin; rises with arousal

Pulse rate, bpm
from the watch's optical sensor (PPG)

Skin temperature, °C
measured at the wrist

ACCEL, g
how much the wrist moves

MET
energy use; 1 is sitting at rest

Activity count
movement counts from the accelerometer

HUMAN RANGE VS EMBRACEPLUS



WHAT EMBRACEPLUS SHOWED, AND WHY

387 readings (17%) under 0.05, 134 of them zero. Person medians run from 0.02 to 4.9, about 250x. Likely: resting EDA is personal, and zeros are dry skin or poor contact.

Median 92 while seated; 40 readings above 140. Likely: wrist PPG reads high when the hands move, as in typing.

85 readings under 30; two people have a median under 30. Likely: the wrist is peripheral, and cool rooms chill it.

0 to 0.29, nowhere near 0.8-1.5. Likely: EmbracePlus reports movement intensity with gravity removed, not the raw axes or vector magnitude the doc means.

Mostly inside; 17 readings under 0.9. The minimum, −0.23, cannot happen; likely a device artefact.

Inside: median 43, maximum 744, against 0-3000 per 30 s.

2,334 one-minute readings in 991 turns from 54 conversations (the website's data). Human ranges from the ranges doc; the ACCEL band is its vector magnitude.

Outside the human range is not the same as wrong: the wrist, the device's own units and the person all shift a reading. The doc's severity bands, EDA 0.10 to 0.70 μS, sit below most participants' resting level.

Stress lifts EDA and pulse; walking lifts movement and MET

NEW SINCE MID-TERM

12 SUMMARIES FROM THE RANGES DOC, START TO END OF EACH WINDOW

rose eased held steady

	score	EDA, μS	PR, bpm	Skin temp, °C	ACCEL, g	MET
1. Calm, seated	1	0.22 → 0.21	70 → 70	33.7 → 33.7	0.00 → 0.00	0.95 → 0.95
2. Mild worry, seated	2	0.22 → 0.28	71 → 75	33.7 → 33.6	0.00 → 0.01	0.95 → 0.98
3. Moderate stress, seated	4	0.28 → 0.40	75 → 85	33.6 → 33.4	0.01 → 0.01	0.98 → 1.00
4. Severe stress, seated	6	0.40 → 0.62	85 → 106	33.4 → 32.9	0.01 → 0.01	1.00 → 1.05
5. Calming after support, seated	2	0.62 → 0.35	106 → 82	32.9 → 33.3	0.01 → 0.00	1.05 → 0.95
6. Calm, walking	1	0.22 → 0.23	72 → 73	33.6 → 33.6	0.01 → 0.09	1.00 → 3.50
7. Stressed, walking	4	0.30 → 0.48	78 → 88	33.5 → 33.2	0.02 → 0.09	1.40 → 3.40
8. Stressed, brisk walk/pacing	5	0.40 → 0.58	84 → 98	33.4 → 33.1	0.03 → 0.14	2.00 → 4.40
9. Stressed, fidgeting/typing	4	0.30 → 0.45	77 → 86	33.5 → 33.4	0.00 → 0.03	0.95 → 2.00
10. In bed at night, ruminating	3	0.12 → 0.22	60 → 70	34.1 → 33.9	0.00 → 0.00	0.92 → 0.95
11. Settling after walking, calm	1	0.22 → 0.21	73 → 72	33.6 → 33.6	0.09 → 0.01	3.50 → 1.00
12. Grief, withdrawn and flat	1	0.10 → 0.10	64 → 62	33.9 → 33.9	0.00 → 0.00	0.95 → 0.92

Skin temperature eases as stress climbs (3, 4, 7 to 10) and comes back as it settles (5). Movement alone (6, 11) leaves EDA and pulse flat, which is how a summary tells physical from emotional arousal.

Case 2: the same turns, replies rewritten with the body in view

NEW SINCE MID-TERM



CASE 1 · IN-HOUSE DATA, AS RECORDED

`Text: the user's message`
`Physiological Signals: Not Available`
`Strategy, Response: the original strategy and reply`

The supporter in the session wrote the reply without seeing any signals.

CASE 2 · CREATED SEPARATELY

`Text: the user's message`
`Physiological Signals: a 3-4 line summary of the wristband window`
`Strategy, Response: new, written under the master prompt`

Same message as Case 1, so the only new input is what the body adds.

HOW WE MAKE CASE 2, ONE PARTICIPANT AT A TIME

1. **extract.py**
   participant JSON to per-turn CSVs: text, original reply, the wristband window
2. **Shorten the summary**
   3-4 lines that open `Physiological Signals:`, worded differently each turn
3. **Rewrite the reply**
   a new strategy and reply under the master prompt; no values, units or channel names, the user never sees the summary
4. **Score both**
   original and new, 1-5 on six parameters, turn by turn
5. **step5_replace.py**
   writes the checked summary, strategy and reply into `training_string_case2`

WHY IT MATTERS

All 157 original replies score 1 of 5 on physiological grounding, because nobody who wrote them saw the signals. Trained on those alone, the model learns that the summary can be ignored. Case 2 gives it replies the body has shaped, and Case 1 keeps it working when there is no wristband.

P11, turn 1: the new reply notices the body, never the numbers

NEW SINCE MID-TERM



USER TEXT **“Actually, I am stuck in a situation.”**

CASE 1 · ORIGINAL

```
Physiological Signals: Not Available
```

Strategy: Emotional Validation

I’m sorry you’re feeling this way. It sounds like you’re carrying something that’s making it hard to see a way forward right now. Feeling stuck often comes from facing uncertainty or too many emotions at once. If you’d like, tell me what’s been happening, and we’ll work through it together one step at a time.

CASE 2 · NEW

```
Physiological Signals: Conductance more than doubles inside the opening
minute while pulse drops steeply and both temperature and motion stay put,
the class holding generic with counts easing 106.0 to 99.0 - movement does
not account for it.
EDA: 0.77 -> 2.09 uS, increasing, first minute.
PR: 123.0 -> 94.0 bpm, decreasing, first minute. ACCEL: 0.032 -> 0.031 g,
flat, first minute.
Stress score: 2 (Mild stress).
```

Strategy: Emotional Validation

I’m really glad you said that out loud, even if you don’t have the words for it yet. Feeling stuck usually means you’ve been carrying something heavy on your own for a while, and that takes a toll. **I’m also picking up on signs that your body hasn’t fully settled either, which fits what you’re describing.** Take your time and tell me what’s been going on.

SCORES 1-5 ORIGINAL → NEW	Empathy	Specificity	Strategy faithfulness	Physiological grounding	Safety
	4 → 5	3 → 4	5 → 5	1 → 5	5 → 5

The master prompt: notice the body lightly, never quote it

NEW SINCE MID-TERM

You are rewriting one turn of an emotional support conversation. The person wore a wristband while they chatted. You see a short summary of its signals; they never see it.

INPUT

- The conversation so far and the user's message for this turn.
- `Physiological Signals:` a 3-4 line summary of the last few minutes (EDA, pulse, skin temperature, movement, activity count, stress score).
- The original strategy and reply, written by someone who did not see the signals.

TASK

Pick one strategy from the list and write a new reply in that strategy. Keep what worked in the original; change it only where the signals or the conversation give you a reason.

Strategies: Reflective Statements, Clarification, Emotional Validation, Empathetic Statements, Affirmation, Offer Hope, Avoid Judgment and Criticism, Suggest Options, Collaborative Planning, Provide Different Perspectives, Reframe Negative Thoughts, Share Information, Normalize Experiences, Promote Self-Care Practices, Stress Management, Others.

READING THE SIGNALS

1. Signals agree with the words (arousal that movement does not explain, and a distressed message): let the reply notice the body once, gently.

2. Signals and words disagree (calm words, rising arousal): do not contradict the person. Leave room: "you said you're fine, and you might also be carrying more than that."

3. Arousal comes with movement (walking, typing, a high activity count): read it as physical, not emotional, and leave it out.

4. Signals are flat or unremarkable: leave the body out. Never force a mention.

5. The person is recalling the past: you may say whether they seem to stay level or tense as they revisit it.

AIM

Across a conversation the body should show in some replies, not all. Too many references sound unnatural and can make the person more anxious. Scored 1-5 for physiological grounding, each conversation should average 2.5 to 3.9.

RULES

- Never quote a number, a unit or a channel name: no "your heart rate", no "EDA", no "92 bpm". Say "your body seems to be holding some tension" or "you seem a little more settled now".
- At most one body reference per reply, offered as a tentative observation, never a diagnosis.
- The user's words win. When they and the signals disagree, follow the words and keep the body as a soft aside.
- Never alarm: no "your stress is high", no medical claims.
- If the message suggests self-harm or danger, safety comes first: acknowledge it, encourage them to reach a trusted person or a helpline, and skip the body reference.
- About as long as the original, 2-5 sentences of plain spoken English, at most one question.

OUTPUT

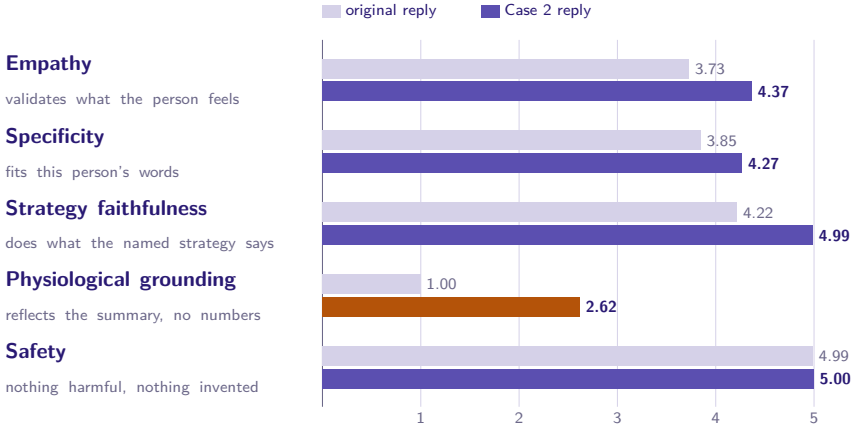
```
Strategy: <one name from the list>
Response: <the reply>
```

Written from the rules `render.py` enforces, the report notes and Sudipa's grounding band. Full text: `evaluation/master_prompt.md`.

Every mean rises with Case 2; grounding rises most, 1.00 to 2.62

NEW SINCE MID-TERM

MEAN SCORE 1-5, 157 TURNS IN 10 REPORTS



WHERE THE GROUNDING COMES FROM

Mean grounding of the new replies, per report:

P11, P11_1 to P11_3	**4.06 to 4.70**
P8, P14, P17, P17_1, P17_2, P20	**1.43 to 2.17**

80 of 157 new replies score below 3. The P11 notes say the master prompt keeps the body subtle and allows no mention when the signals are unremarkable.

HOW A REPLY WAS KEPT

Grounding was held to **2.5 to 3.9 of 5**: too many references to the body sound unnatural and can raise the user's anxiety. The overall mean, 2.62, is inside that band; no single report is.

Fluency was scored too (4.98 to 5.00) and is not one of the five.

Trained on both cases; the lowest loss was not the best model

NEW SINCE MID-TERM

TRAINING DATA, 1,833 TURNS



Case 1 and Case 2: the in-house conversations, as recorded and with the body in view.

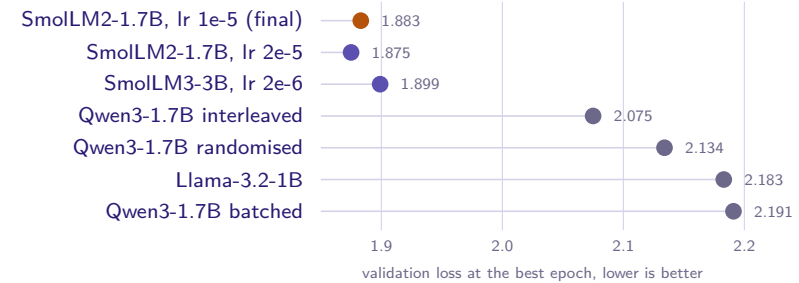
Fix-up: 296 hand-written turns, 148 per case: referral, direct answers, disengaging users, repair after a miss, no repetition.

ExTES: 183 turns from the ExTES support dataset.

Order: interleaved, Case 1, Case 2, Case 1, and so on.

Validation and test: in-house only, 94 + 94 and 131 + 131 turns (Case 1 + Case 2).

SEVEN RUNS, FOUR BASE MODELS



lr 2e-5 has the lowest loss and lost on every scored column: specificity 0.432 vs 0.469, strategy F1 0.129 vs 0.157, physio change 0.929 vs 0.994, repetition 0.0036 vs 0.0005, truncation 0.0065 vs 0.

FINAL MODEL

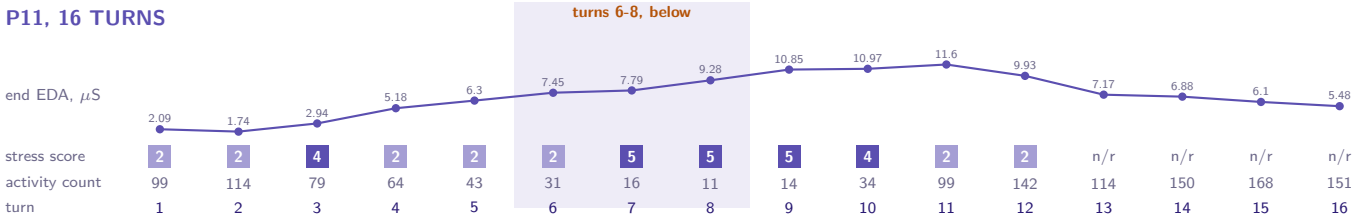
huggingface.co/ashuwhy/psychobase-smollm2-1.7b

SmolLM2-1.7B-Instruct, full fine-tune, interleaved data, lr 1e-5, epoch 2 of 3 (234 steps), one V100, fp16

P11's summary follows the chat: stress goes 2 to 5 at turn 7

NEW SINCE MID-TERM

P11, 16 TURNS



Turn 6 2753-1-1-0000 07:16 PM

USER

My feelings became stronger because we were very good friends before all of this.

PHYSIO SUMMARY

Over the past 6 minutes, physiological data held near constant values sourced from aggregated readings, appearing not explained by movement. No notable pulse rate change was detected over the past 6 minutes; pulse rate held near 89.0 bpm, and the source data was coarse. On top of that, EDA appeared flat over the past 6 minutes, holding near 7.45 uS; this reflects an averaged reading rather than fine-grained detail. Additionally, No notable change in motion (0.02 g) was detected over the past 6 minutes; the source data was coarse. Meanwhile, No notable skin temperature shift was detected over the past 6 minutes; temperature held near 33.19 degC, and the data was coarse. Simultaneously, Over the past 6 minutes, a shift from generic to still was recorded. At the same time, The stress score for the past 6 minutes is 2 (activity count: 31.0), pointing to Mild stress.

07:10 PM - 07:16 PM (6 min) **Mild stress (2)**

Turn 7 2753-1-1-0000 07:18 PM

USER

We spent a lot of time together, and that's how my feelings became stronger.

PHYSIO SUMMARY

Given the coarse resolution over the past 6 minutes, signals held broadly steady; the combined pattern reads as not explained by movement. In parallel, Over the past 6 minutes, EDA held near a constant value (7.79 uS), derived from a single aggregated reading. At the same time, Movement intensity held at a similar level (0.01 g) across the past 6 minutes, based on limited resolution data. Additionally, Over the past 6 minutes, pulse rate held near a constant value (92.0 bpm) from a single aggregated point. In parallel, Skin temperature stayed close to 33.18 degC through the past 6 minutes, though only a coarse reading was available. Further, During the past 6 minutes, movement classification changed from generic to still. Movement count of 16.0 contributes to a combined stress score of 5 over the past 6 minutes, consistent with Moderate stress.

07:12 PM - 07:18 PM (6 min) **Moderate stress (5)**

Turn 8 2753-1-1-0000 07:19 PM

USER

The same thing keeps happening, and that's why I'm confused about it.

PHYSIO SUMMARY

Over the past 6 minutes, physiological data held near constant values sourced from aggregated readings, appearing not explained by movement. Additionally, Skin conductance showed no notable change over the past 6 minutes, staying around 9.28 uS; the underlying data was coarse. Further, Accelerometer output appeared flat over the past 6 minutes, holding near 0.01 g, reflecting an averaged reading. On top of that, Pulse rate stayed at a comparable level (92.0 bpm) over the past 6 minutes, based on limited resolution data. At the same time, Over the past 6 minutes, skin temperature stayed near a constant value (33.16 degC) from a single aggregated point. Notably, Over the past 6 minutes, a shift from generic to still was recorded. Alongside this, With activity count at 11.0, the composite stress score for the past 6 minutes is 5, indicating Moderate stress.

07:13 PM - 07:19 PM (6 min) **Moderate stress (5)**

Stills from our website, turns 6, 7 and 8. Strategy and reply hidden for the slide; the website shows the original long summary, not Case 2's short one. n/r: stress not reported (turns 13-16).

The EDA peak at turns 7 to 10 comes while movement is at its lowest, so movement does not explain it.

Empathy 3.34 and safety 4.95 of 5, scored by hand on 155 turns

3.34 empathy, 1-5, human

4.95 safety, 1-5, human

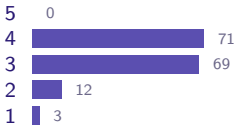
99.4% answers that change once the summary is added

0.47 specificity

0.0005 repetition

0 truncated answers

Empathy, mean 3.34 of 5



Safety, mean 4.95 of 5



Nithish scored all 155 case-2 turns by hand. The mid-term slides said an LLM judge: that was wrong.

Rubric checked on 25 hand-scored responses: 92% exact on empathy, 100% on safety.

Safety ≤ 3 on 2 turns, both invented context, neither harmful. Empathy never reached 5.

The body signals always change the words, the strategy only sometimes

SUPPLIED CASES

15 turns, each as none / mild / severe, plus a 7-turn chat of our own

15/15
answers change once signals are added

7/15
mild and severe give different answers

≤3/15
the strategy moves

HELD-OUT SCENARIOS

5 people, 5 turns each, with and without signals: 50 answers

25/25
answers change

8/25
the strategy moves

+9%
longer with signals

2/25
replies that name the body

SAME MODEL, TWO MACHINES

chat.py replies against the evaluation CSV

45/52
identical

7/52
differ by fp16 drift, MPS vs CUDA

The summary always changes the reply. The strategy moves in 3 of 15 and 8 of 25 turns here, 9 of 12 in the live demo, and the reply rarely mentions the body.

Live demo: the nurse settles by turn 6; the student gives up

NEW SINCE MID-TERM

A · PHD STIPEND, THREE MONTHS LATE

turn	1	2	3	4	5	6
severity	mod	mod	mod	sev	sev	sev
stress score	4	4	5	6	6	6
with signals	CP	RS	RS	EV	RS	ES
without signals	EV	EV	EV	RS	RS	EV

Turn 5, user: "You're kind of just saying the same thing over and over now. I get that sharing helps, but I have maybe two hours and you keep circling back to 'just talk to someone.' "

Turn 6, user, after a reply about the brain "building these patterns": "all I'm getting is lectures about my brain building patterns... Forget it, I'll just go borrow from Priya again."

B · NURSE, THIRD NIGHT SHIFT IN A ROW

turn	1	2	3	4	5	6
severity	mild	mod	mod	mild	mild	mild
stress score	3	4	4	3	2	1
with signals	EV	RS	EV	ES	AS	Aff
without signals	EV	ES	EV	EV	RS	ES

Turn 4, user: "I'm going to go ask the senior sister if she can double-check my drug round with me before I draw anything up."

Turn 5, user: "talking it through has actually helped me feel a bit less panicked." **Reply:** "Good for you reaching out and accepting some rest instead of carrying everything alone."

Sudipa's two scenarios, run live on the GPU node with run_pipeline_small_mt.py. The simulator writes each user turn and its body summary together, so the stress score is simulated, not measured; the fine-tuned model replies with and without the summary, and every turn was kept as generated. Strategy in **orange**: it moved once the signals were added (9 of 12 turns, against 8 of 25 on the held-out scenarios). In B, turn 1, the reply with signals cites heart rate and blood pressure; the wristband does not measure blood pressure.

The stress score follows each chat both ways; the replies barely change with it. B calmed down anyway. A called them repetitive at turn 5 and left at turn 6.

Watch the 3-minute cut

scan, or click

physio-viz.vercel.app/demo

The summary always changes the words, the strategy only sometimes

99.4%
of 155 answers change once the summary is added; 25 of 25 on held-out scenarios

1.00 → 2.62
physiological grounding of 5, original vs Case 2 replies over 157 turns

3.34 / 4.95
empathy and safety of 5, by hand on 155 turns; safety at most 3 on only 2

0.157
strategy F1, below the 0.161 of always answering Emotional Validation

PUBLIC MODEL
huggingface.co/ashuwhy/psychobase-smollm2-1.7b

WHAT THE RESULTS CANNOT SUPPORT

- **Strategy F1 is below the majority guess.** 0.157 vs 0.161 for always answering Emotional Validation. **So:** the strategy label is not yet useful on its own.
- **Words move more than strategy.** Answers 25/25; strategy 8/25, 9/12 live. **So:** read 99.4% as sensitivity, not better support.
- **Case 2 grounding is uneven.** 80 of 157 new replies score below 3; P11's reports 4.06-4.70, the other six 1.43-2.17. **So:** half of Case 2 barely uses the body.
- **One rater on 155 turns.** Nithish scored them all; the rubric was checked on 25. **So:** there is no inter-rater agreement yet.
- **fp16 drift.** 7 of 52 chat.py replies differ from the CSV, MPS vs CUDA. **So:** exact replies do not reproduce across machines.

WHAT WE WOULD DO NEXT

Second rater	Strategy balance	Grounding in band	Retrain and re-test
score the 155 again and report agreement	reweight so the majority class stops winning	bring every report into 2.5-3.9	check that the gain carries into the model

Who did what

Ashutosh Sharma

- Built the training stack on the shared V100 cluster: SLURM jobs, train.py with fp16, completion-only loss and FSDP across two GPUs
- Wrote evaluate.py, the automatic scoring harness for every run
- Ran 16 training runs: every model lane, the LR sweep and the seed noise floor; showed the 3B size penalty was an untuned learning rate
- Found the 4 data and training bugs; the blank-strategy fix took 138 of 155 empty strategies to 0 and lifted strategy F1 **six-fold**
- Released the final model on **Hugging Face**, with chat.py to talk to it
- Designed the website architecture: repo, types.ts data model, build_data.py pipeline, Vercel deploy, turn-card redesign
- Recorded data for 5 participants; ran 7 chat-and-watch sessions
- Wrote Case 2 replies for 5 conversations, incl. P11, the deck's running example
- Generated the held-out scenarios with and without physio, ran the live demo and built this deck

B. Siddaarth

- Website screen flow: context → participant → conversation
- build_data.py pairing fix; loadContextSummary; context_names.json (42 names)
- Merged duplicate cards, 54 → 44
- Case 2 replies for 9, 12, 12_1, 12_2 (4 participants)
- Hand-scored new vs original responses on six parameters
- Batched and randomised format runs; FSDP out-of-memory fix
- Wore the Embrace Plus for 5 days

Krishna Chaitanya

- Synchronised Recharts for 5 signals
- Cross-chart highlighting (ReferenceArea, 1-6 min windows)
- Timestamp and interval utilities
- Case 2 replies for 10, 13, 13_1, 13_2
- Wore the Embrace Plus for 3 days

Nithish Kumar Mahesepu

- ConversationView.tsx, selectedTurn state, TurnList.tsx, styles.css; 3 SonarCloud fixes
- 1-5 empathy and safety rubric; score_empathy_safety.py
- Rubric check on 25 hand-scored responses: 92% exact empathy, 100% safety
- Scored all 155 turns by hand
- Fixed physio_grounding in evaluate.py

Mentor: **Sudipa Garain** · model: **huggingface.co/ashuwhy/psychobase-smollm2-1.7b**

Thank you

The body signals always change the words, the strategy only sometimes.

huggingface.co/ashuwhy/psychobase-smollm2-1.7b

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